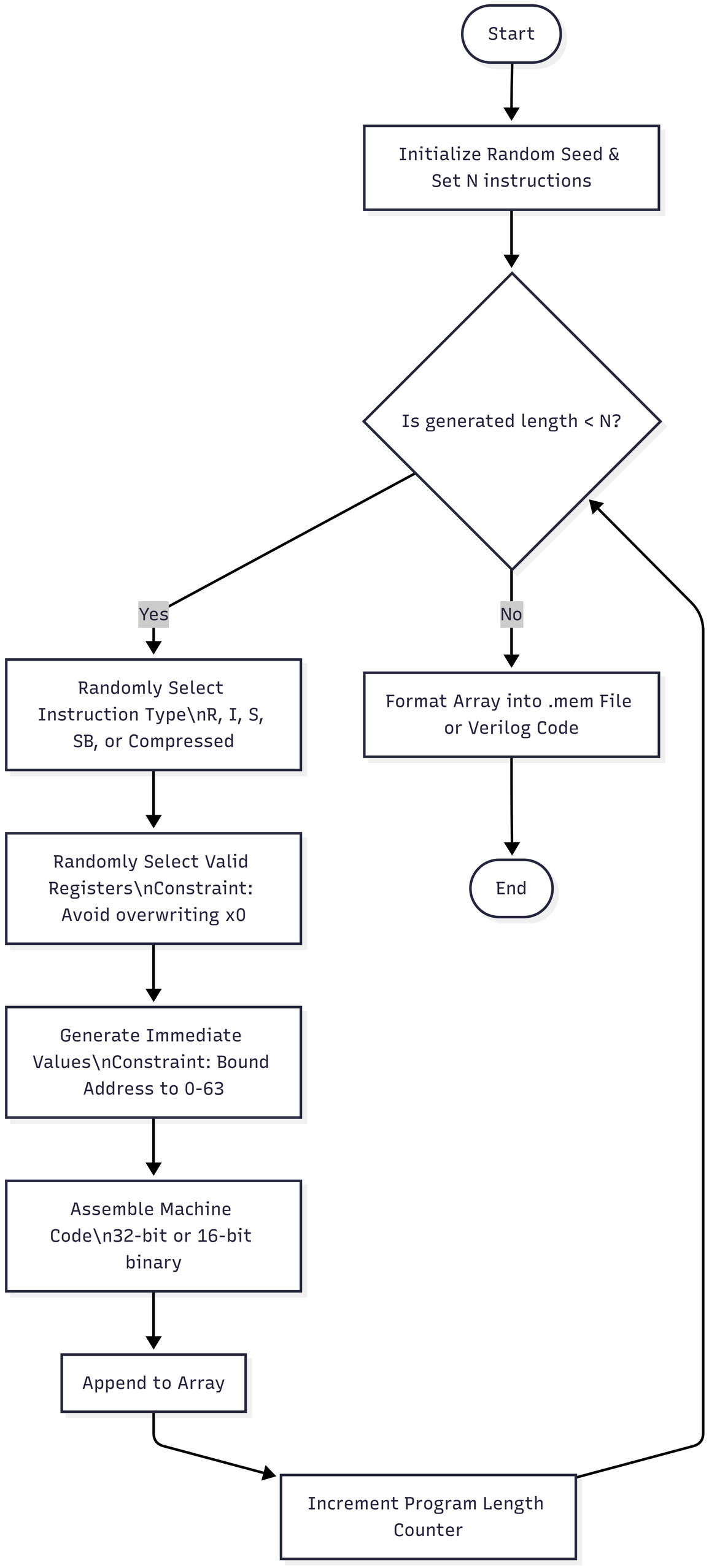
Start
Initialize Random Seed & Set N instructions
Is generated length < N?
Yes
No
Randomly Select Instruction Type\nR, I, S, SB, or Compressed
Randomly Select Valid Registers\nConstraint: Avoid overwriting x0
Generate Immediate Values\nConstraint: Bound Address to 0-63
Assemble Machine Code\n32-bit or 16-bit binary
Append to Array
Increment Program Length Counter
Format Array into .mem File or Verilog Code
End